

ABHISHEK TYAGI

Lead AI/ML Engineer | Generative AI, RAG, LLMops and MLOps

Open to relocation across Germany (Munich, Berlin, Erlangen, Hamburg, Stuttgart, Frankfurt) | Currently Gurugram, India
8744809165 | abhishektyagi101@gmail.com | linkedin.com/in/abhishek-tyagi-101 | github.com/abhishektyagi101

Work authorisation: EU Blue Card eligible (German Skilled Immigration Act) | Accelerated skilled-worker procedure (§81a AufenthG) ready | Holder of a current German-issued 2-year multi-entry Schengen visa, third consecutive Schengen visa after Spain and Austria | Relocation-ready, no employer risk on timeline

PROFESSIONAL SUMMARY

Lead AI/ML Engineer with 11+ years in production software and machine learning, the last four building and operating Generative AI systems for a German-headquartered industrial group. Shipped a production Retrieval-Augmented Generation platform on Azure OpenAI and AWS, built real-time anomaly detection for gas turbines, and established the MLOps and LLMops practice around both: model and prompt versioning, evaluation gates on every release, drift, latency and cost monitoring. Work daily with Siemens Energy colleagues in Germany, Spain and Austria in an Agile/Scrum and SAFe PI Planning setup. Relocating permanently to Germany to own an AI platform or Generative AI product line.

AVAILABILITY AND RELOCATION TO GERMANY

- **Immigration:** Meets all EU Blue Card criteria (recognised engineering degree, salary threshold, IT-professional route). Employer may also use the accelerated skilled-worker procedure (§81a AufenthG) via the local Ausländerbehörde.
- **Timeline:** 90-day notice period runs in parallel with the visa process, so the standard 4–6 week issuance completes inside it. Start date: 3 months from signed contract, with no additional waiting time for the employer.
- **Travel today:** Valid Indian passport and an active **German-issued 2-year multi-entry Schengen visa** (90 days per stay), so on-site interviews, team days and an early onboarding visit in Germany are possible at short notice.
- **Schengen visa history:** Third consecutive multi-entry Schengen visa — previously issued by Spain and by Austria, now by Germany. Clean immigration record with repeated business travel to the Schengen area and full compliance with every stay.
- **Germany track record:** Four years of delivery with Siemens Energy Germany. Most recently on site in **Berlin (1–3 September 2026) as Project Lead at the DBPD PI Planning event**, plus on-site work in Duisburg and European assignments in Barcelona and Vienna.

TECHNICAL SKILLS

Generative AI & LLM: Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Azure OpenAI, GPT-4, AWS Bedrock, LangChain, LlamaIndex, Hugging Face Transformers, Model Context Protocol (MCP), prompt engineering, embeddings, semantic search, hybrid retrieval, re-ranking, chunking strategies, groundedness and citation enforcement, guardrails, LLM evaluation, fine-tuning

Vector Databases: AWS OpenSearch, Elasticsearch, FAISS, Qdrant, Pinecone, Weaviate, response caching

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, XGBoost, SVR, stacked ensembles, time-series forecasting, anomaly detection, predictive maintenance, NLP and text classification, computer vision, feature engineering, model optimisation

MLOps & LLMops: MLflow (experiment tracking, model registry), model and prompt versioning, CI/CD for models, automated retraining, canary and A/B rollout, offline and online evaluation, drift detection, cost and token monitoring, rollback, reproducibility and lineage

Cloud & Platform: AWS (SageMaker, Bedrock, Lambda, Step Functions, EventBridge, API Gateway, OpenSearch, CloudWatch, SNS, SQS, S3, IAM, QuickSight), Microsoft Azure (Azure OpenAI, Azure ML), Google Cloud Platform, Kubernetes, Docker, Terraform, FastAPI, REST API design, microservices, event-driven architecture, versioned inference endpoints, asynchronous batch inference, autoscaling

DevOps & Quality: GitLab CI/CD, Git, GitHub Actions, pytest, unit and integration testing, SonarQube security gates, code review, Prometheus, Grafana, OpenTelemetry, incident response and postmortems

Programming & Data: Python, C#/.NET, Java, SQL, PostgreSQL, MongoDB, Snowflake, Pandas, NumPy, Spark basics, data ingestion and transformation pipelines

Compliance, Governance & Leadership: GDPR / DSGVO-aligned data handling, EU AI Act awareness, role-based access control, secrets management, audit trails, EU data residency, technical leadership, solution architecture, system design, mentoring, distributed international teams, stakeholder management, Agile / Scrum, SAFe PI Planning, roadmap and capacity planning

Domain: Industrial IoT, Industrie 4.0, gas turbines and industrial compressors, condition monitoring, sensor time-series data, energy and power generation, manufacturing analytics

PROFESSIONAL EXPERIENCE

Lead AI/ML Engineer (Senior Solution Expert – IT Application)

Siemens Energy Industrial Turbomachinery Pvt. Ltd. (Siemens Energy Group, HQ Germany), Gurugram, India | Feb 2022 – Present

- Lead Generative AI and predictive-maintenance initiatives on Azure OpenAI and AWS for gas compressor optimisation, taking solutions from proof of concept through to production deployment and ongoing operation.
- Architected and shipped a production RAG platform using LLMs and Model Context Protocol (MCP) for context-aware retrieval across technical documentation and industrial sensor data. Used daily by engineering and service teams at three international sites; document lookups that previously took 20–30 minutes now return an answer in under a minute.
- Set up the LLMops practice from scratch: prompt and model versioning, MLflow experiment tracking and model registry, a golden evaluation set gating every release, and human review of sampled production answers. Ungrounded answers fell roughly 40% across three release cycles.
- Built a real-time anomaly detection system for gas turbines on AWS Step Functions, EventBridge, Lambda and SageMaker with automated SNS/SQS alerting, replacing manual condition-monitoring checks and surfacing developing faults days earlier.
- Own the architecture for secure, scalable ML APIs: authentication, per-client rate limiting, versioned SageMaker endpoints behind API Gateway, autoscaling, an asynchronous batch inference path on Lambda, and CloudWatch alerting that pages on-call when latency regresses. Cut average cost per LLM request by roughly 35% through response caching, request batching and routing simpler queries to smaller models instead of sending everything to GPT-4.
- Developed time-series models (SVR, XGBoost, stacked ensembles) for turbomachinery residual analysis across a Python / C++ / .NET stack, improving accuracy over the production baseline.

- Handle governance for a platform holding customer equipment data: role-based access control, secrets managed in AWS, EU data-residency constraints, and audit trails covering who asked what and which documents an answer used.
- Lead software and DevOps engineers across India, Germany and Spain through the full SDLC in an Agile/Scrum and SAFe environment; enforce quality through design review, SonarQube security gates and GitLab CI/CD.
- Represent the AI/ML workstream as **Project Lead at Siemens Energy DBPD PI Planning, on site in Berlin (1–3 September 2026)**: committed team capacity, negotiated cross-team dependencies with German product and platform stakeholders, and agreed the AI roadmap for the coming increment.

Tech: Python, Azure OpenAI, GPT-4, LangChain, MCP, AWS (SageMaker, Lambda, Step Functions, EventBridge, OpenSearch, SNS/SQS), PyTorch, Scikit-Learn, XGBoost, MLflow, C#/.NET, Docker, Kubernetes, GitLab CI/CD, SonarQube

Senior Software Engineer (Technical Lead)

BlackBox IP Corporation, Noida, India | Feb 2020 – Feb 2022

- Technical lead for a patent IDS analytics platform; designed pattern-recognition and NLP pipelines that cleaned, normalised and classified several hundred thousand patent documents.
- Replaced a manual analyst review step with text classification and predictive models, reducing per-document handling from several minutes to a few seconds.
- Delivered executive analytics dashboards on AWS QuickSight for enterprise IP clients; introduced code review and unit-testing standards and mentored junior engineers.

Tech: Python, Scikit-Learn, NLP, AWS QuickSight, SQL, Git

Associate Consultant

Capgemini Technology Services India Ltd., Noida, India | Oct 2017 – Feb 2020

- Engineered and deployed machine learning models on AWS for data-security use cases, and owned the ingestion side: large-scale web scraping, text extraction and NLP cleaning pipelines feeding model training.
- Restructured a sprawling Python codebase into modular components using object-oriented design patterns, removing 30% of duplicated code.

Tech: Python, AWS, Scikit-Learn, Beautiful Soup, SQL

Associate

Wipro Limited, Noida, India | Sep 2015 – Jul 2017

- Automated network and operating-system diagnostics in object-oriented Python, then analysed the resulting infrastructure telemetry to identify recurring failure patterns and drive permanent fixes.

Technical Support Engineer

IEnergizer IT Services Pvt. Ltd., Noida, India | Oct 2013 – Jun 2014

- Applied Python-based data analytics to diagnose network infrastructure anomalies and provided escalation-level technical troubleshooting.

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Communication Engineering

Hindustan College of Science and Technology, Mathura (GBTU / AKTU), India | 2009 – 2012 | First Division (64.57%)

Recognised in the Anabin database; 4-year equivalent engineering qualification for EU Blue Card purposes (lateral entry after 3-year diploma).

Diploma in Electronics Engineering (3-year, full-time)

Government Polytechnic Saharanpur, India | 2006 – 2009

CERTIFICATIONS

- Master Program in Artificial Intelligence Engineer, Simplilearn (Certification Code 24901379)
- TensorFlow Developer, IBM / Simplilearn
- Industrial Compressor Design and Predictive Maintenance, Siemens Energy Global Training Centre, Duisburg, Germany

ADDITIONAL INFORMATION

- **Languages:** English (C1, professional working proficiency); German (A2, actively studying toward B2); Hindi (native).
- **International exposure:** DBPD PI Planning, Berlin, Germany (Sep 2026, as Project Lead); Domain-based PI Planning 2026 (Vienna); Siemens Energy DPS 2025 (Barcelona); Global Training Centre, Duisburg, Germany.
- **Notice period:** 90 days, negotiable / buy-out possible — runs in parallel with the visa process.
- **References and documents:** Degree certificates, employment references and passport available on request.